Sri Lanka Institute of Information Technology



IE1030 - Data Communication Networks
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Network Design Group Project

Proposal Document

WD Group 13

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Introduction

In today's environment having a network of computers to help communication, exchange of information Internet access and the overall efficiency of an organization is essential. A strong network can give employees the ability to share resources communicate and use both wired and wireless devices well.

The project involves designing an enterprise network for a two-story office building that measures 18 m by 10 m. The network is designed to meet the communication and connection needs of employees on both floors. The network will allow connections, for personal devices and will give Internet access through wireless access points placed in appropriate locations.

This network design project uses a router, switches, wireless access points, Cat6 UTP cables, patch panels, outlets, equipment racks and a UPS. The project will follow network wiring techniques and will address link speed, equipment location, cable connections, scalability and cost-effectiveness.

Cisco Packet Tracer will be used to document or design the enterprise topology and the connection of devices. Finally, the project will deliver a topology calculate the number of cables and ports list equipment recommend the location of access points provide an Internet connection and give the cost in Sri Lankan Rupees.

The main purpose of the project is to create an efficient network that will meet the needs of the two-story building while following the basic principles of data communication and network design.

1.Requirement Analysis

1.1 Building Overview and Network Requirements

The proposed network is built for a two-story office building. Each floor has dimensions of 18 m X 10m. The network is meant to offer wired and wireless connections for employees, office devices and mobile users.

The setup includes a Cisco 2911 Integrated Services Router, three 24-port PoE switches, four wireless access points, Cat6 UTP cabling, fiber-optic backbone cabling patch panels, RJ45 outlets, network racks and UPS units.

The goal of this network is to deliver wired and wireless connectivity for employees and visitors. It must also ensure efficient access, to the Internet and organizational resources.

1.2 Ground Floor Requirements and Layout

The ground floor measures 18 m X 10m. It includes 21 wired network connections, made up of 18 desktop PCs and 3 IP phones. There are also 2 wireless access points or APs to support laptops and smartphones with coverage. A separate area is set aside for network equipment. This area holds a TR-GF 12U network rack, a 24-port PoE switch, a 24-port patch panel and an uninterruptible power supply or UPS.

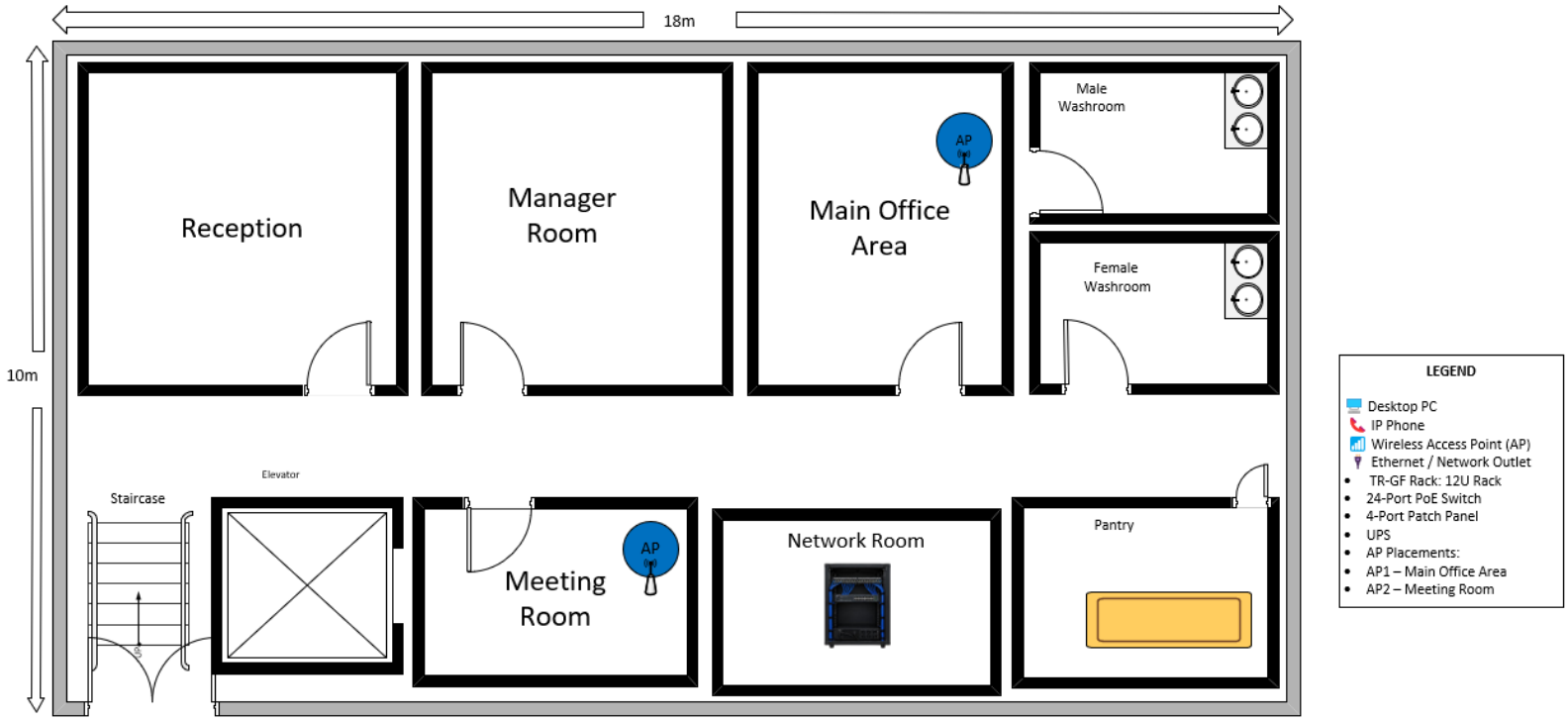
The requirements are listed as follows:

Requirement	Quantity
Floor Area	18m X 10m
Wired Network Drops	21
Desktop PCs	18
IP Phones	3
Wireless APs	2
Network Rack	1 X 12U
24-Port PoE Switch	2
24-Port Patch Panel	1
UPS	1
Horizontal Cabling	Cat6 UTP

All equipment and wireless access point locations have been planned carefully. The goal is to ensure cable paths and solid network coverage throughout the space.

Ground Floor Room by Room Requirements

Room/Area	Desktop PCs	IP Phones	Total Wired Drops	Wireless Clients
Main Office Area	10	1	11	10
Manager Room	2	1	3	2
Reception	4	1	5	4
Meeting Room	2	0	2	10
Total	18	3	21	26



Ground Floor Plan Layout

1.3 First Floor Requirements and Layout

The first-floor measures 18 m X 10m. It has 21 wired network connections and 2 wireless access points. The wired connections are used by 18 desktop PCs and 3 IP phones.

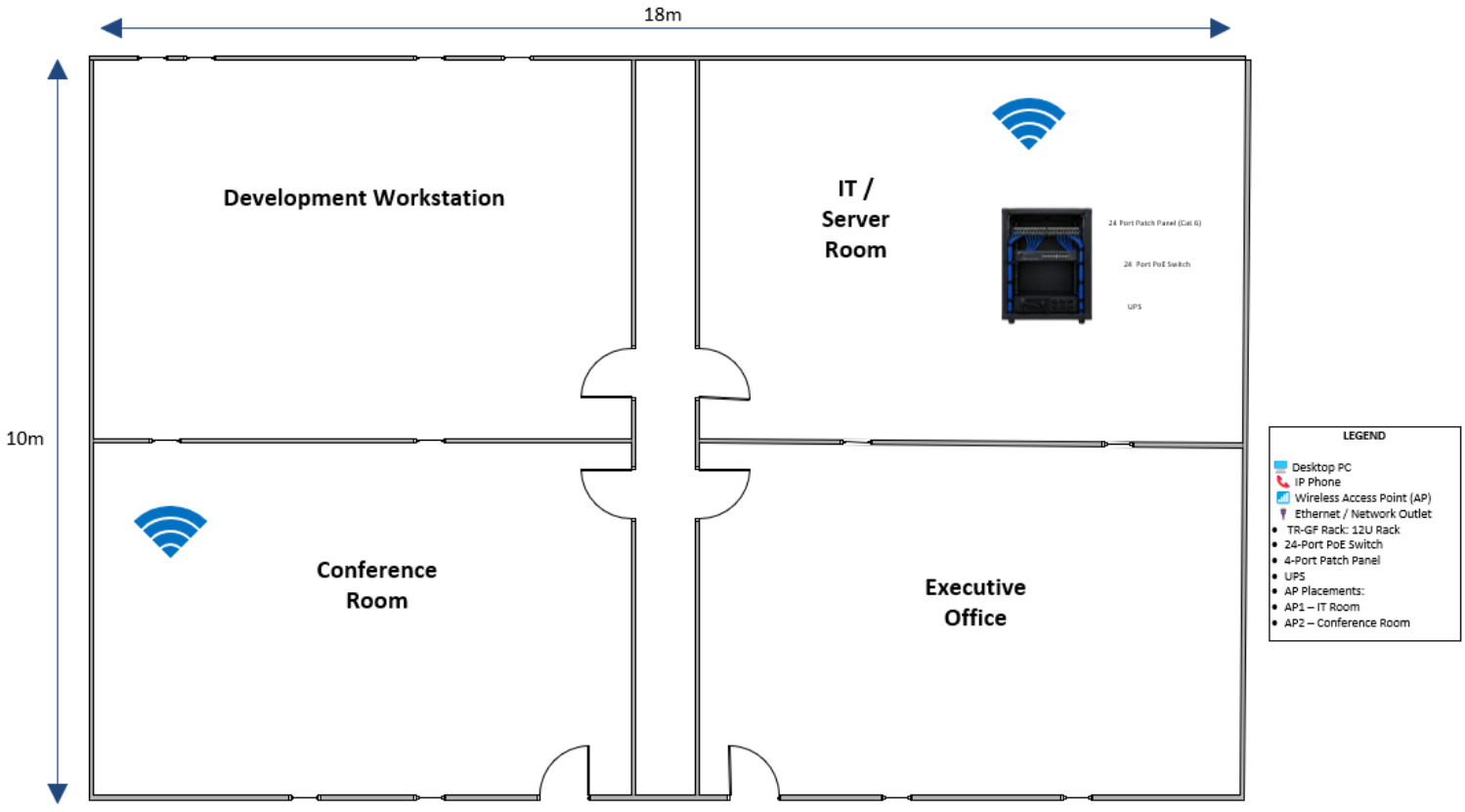
There is a Server/IT Room. This room holds the TR- 12U network rack, a switch, a patch panel and a UPS. The two wireless access points are placed in locations. This ensures wireless coverage across the main working areas of the floor.

The requirements are listed as follows:

Requirement	Quantity
Floor Area	18m X 10m
Wired Network Drops	21
Desktop PCs	18
IP Phones	3
Wireless APs	2
Network Rack	1 X 12U
24-Port PoE Switch	1
24-Port Patch Panel	1
UPS	1
Horizontal Cabling	Cat6 UTP

First Floor Room by Room Requirements

Room/Area	Desktop PCs	IP Phones	Total Wired Drops	Wireless Clients
IT/Server Room	4	1	5	6
Development Workstation	4	1	5	8
Executive Office	2	1	3	6
Conference Room	8	0	8	8
Total	18	3	21	28



First Floor Plan Layout

1.4 Wired and Wireless Device Requirements

The proposed network supports both wired and wireless devices. The wired network provides 42 connections while four wireless APs provide connectivity for 54 wireless client devices.

Wired Requirements

Wired Devices	Ground Floor	First Floor	Total
Desktop PCs	18	18	36
IP Phones	3	3	6
Total Wired Connections	21	21	42

All wired connections use Cat6 UTP cabling. They are connected through the floor-level patch panels. The design target, for wired connections is 1 Gbps Ethernet.

Wireless Requirements

Wireless Devices	Ground Floor	First Floor	Total
Wireless Aps	2	2	4
Laptops	11	14	25
Smartphones/Mobile Devices	15	14	29
Total Wireless Clients	26	28	54

The APs are positioned to provide coverage across the working areas and reduce potential wireless dead zones.

1.5 Internet Connectivity Requirements

The following Internet connectivity options were considered for the proposed office network.

Option	Suitability
Fiber Optic (FTTH/Fiber Broadband)	High Speed and suitable for multiple simultaneous office users
ADSL Broadband	Lower Bandwidth and less suitable for proposed office
4G/LTE Broadband	Suitable as a backup connection

Fiber Optic (FTTH/Fiber Broadband) connectivity is selected as the primary Internet connection because the office requires reliable high-speed connectivity for 42 wired connections and 54 wireless clients.

Requirement	Proposed Specifications
Internet Service Provider	SLT-MOBITEL
Connection Type	Fiber Optic
Service	Fiber Unlimited Pro
Download Speed	500 Mbps
Upload Speed	200 Mbps
Monthly Rental	Rs. 19,900 + Taxes
Internet Gateway	Cisco 2911 Integrated Services Router
Router–Switch Connection	1 Gbps Ethernet
Horizontal Cabling	Cat6 UTP
Inter-Floor Backbone	Multimode Fiber Optic

The selected SLT-MOBITEL Fiber Unlimited Pro package provides 500 Mbps download and 200 Mbps upload speeds with unlimited data. This is suitable for supporting the wired and wireless users and common business applications such as cloud services, video conferencing, file sharing and web access.

The fiber connection terminates at the Optical Network Terminal (ONT). Connects to the Cisco 2911 Integrated Services Router, which acts as the Internet gateway. The router distributes Internet connectivity through the network switches to users on both floors.

Estimated Monthly Data Consumption

The estimated monthly data consumption for the organization is calculated based on the expected number of users and their daily internet usage.

• Total Users/Devices: 42 wired end devices + 54 wireless client devices = 96 active nodes

• Usage per User: 5 GB per user, per day

• Daily Data Consumption: 96 × 5 GB = 480 GB/day

• Monthly Data Consumption: 480 GB × 22 working days = 10,560 GB/month

• Estimated Monthly Requirement: Approximately 10.56 TB/month

2.Logical Network Design

2.1 Network Architecture

The proposed network follows a star-based architecture to give reliable and scalable connectivity throughout the two-story office building. The Internet connection comes from SLT-MOBITEL Fiber is linked to the Cisco 2911 router through the ONT. The router serves as the gateway and connects the network switches.

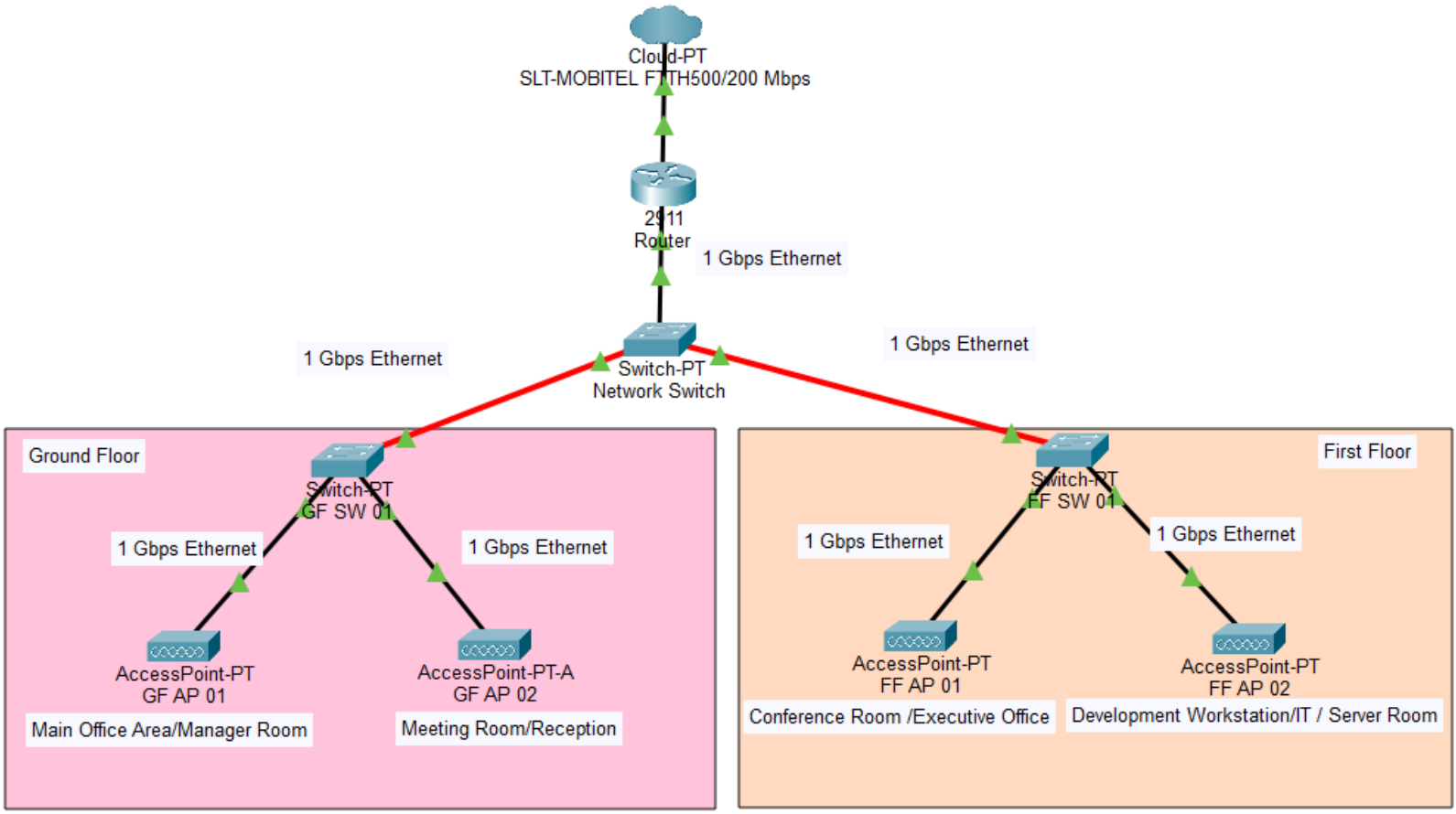
Separate switches are used to provide network connectivity, for the ground floor and the first floor. Wireless access points are attached to the switches to give connectivity for laptops, smartphones and other mobile devices.

The main components of the network are:

Device	Quantity	Function
Cisco 2911 Router	1	Internet gateway and routing
Network Switches	3	Wired network connectivity
Wireless Access Points	4	Wireless connectivity
Ground Floor Network	1	Ground-floor connectivity
First Floor Network	1	First-floor connectivity

2.2 Logical Network Topology

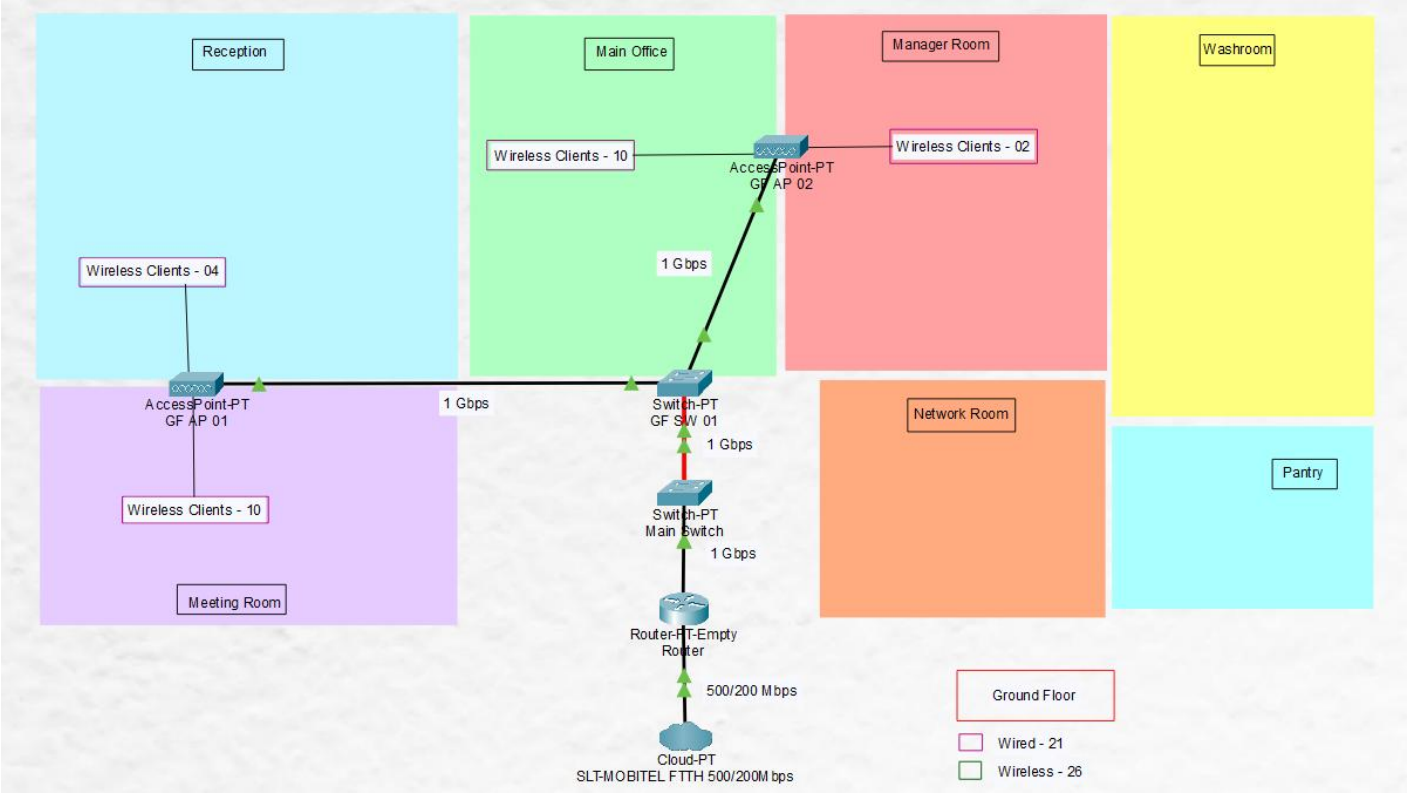
The proposed network uses a **star topology**. In this setup the central switches connect all the network devices, on each floor. A hierarchical design links the router, switches and wireless access points. This design helps with communication. It also makes the network easier to manage. It allows the network to grow easily in the future.



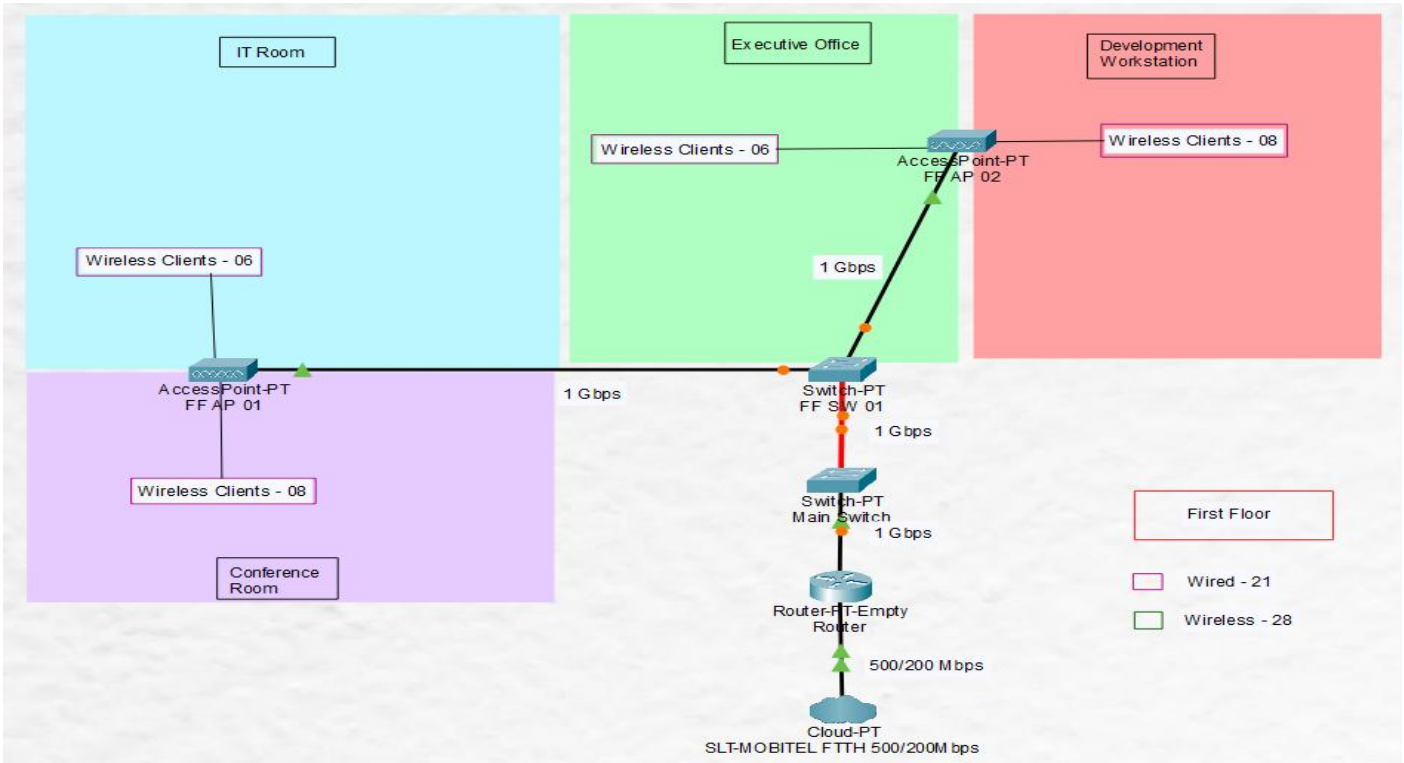
Logical Network Diagram

2.3 Ground and First Floor Network Diagram

Ground Floor Network Diagram



First Floor Network Diagram



2.4 Link Speeds and Connections

Connection	Medium	Speed
ONT → Cisco 2911 Router	FTTH Service	500Mbps down 200Mbps up
Router → Main Switch	Cat6 UTP Ethernet	1 Gbps
Main Switch →Ground Floor Switch	Fiber Optic Backbone	1Gbps
Main Switch → First Floor Switch	Fiber Optic Backbone	1 Gbps
Switch → Wireless AP	Cat6 PoE+ Ethernet	1 Gbps
Switch → Wall outlets/Wired end devices	Cat6 UTP Ethernet	1 Gbps

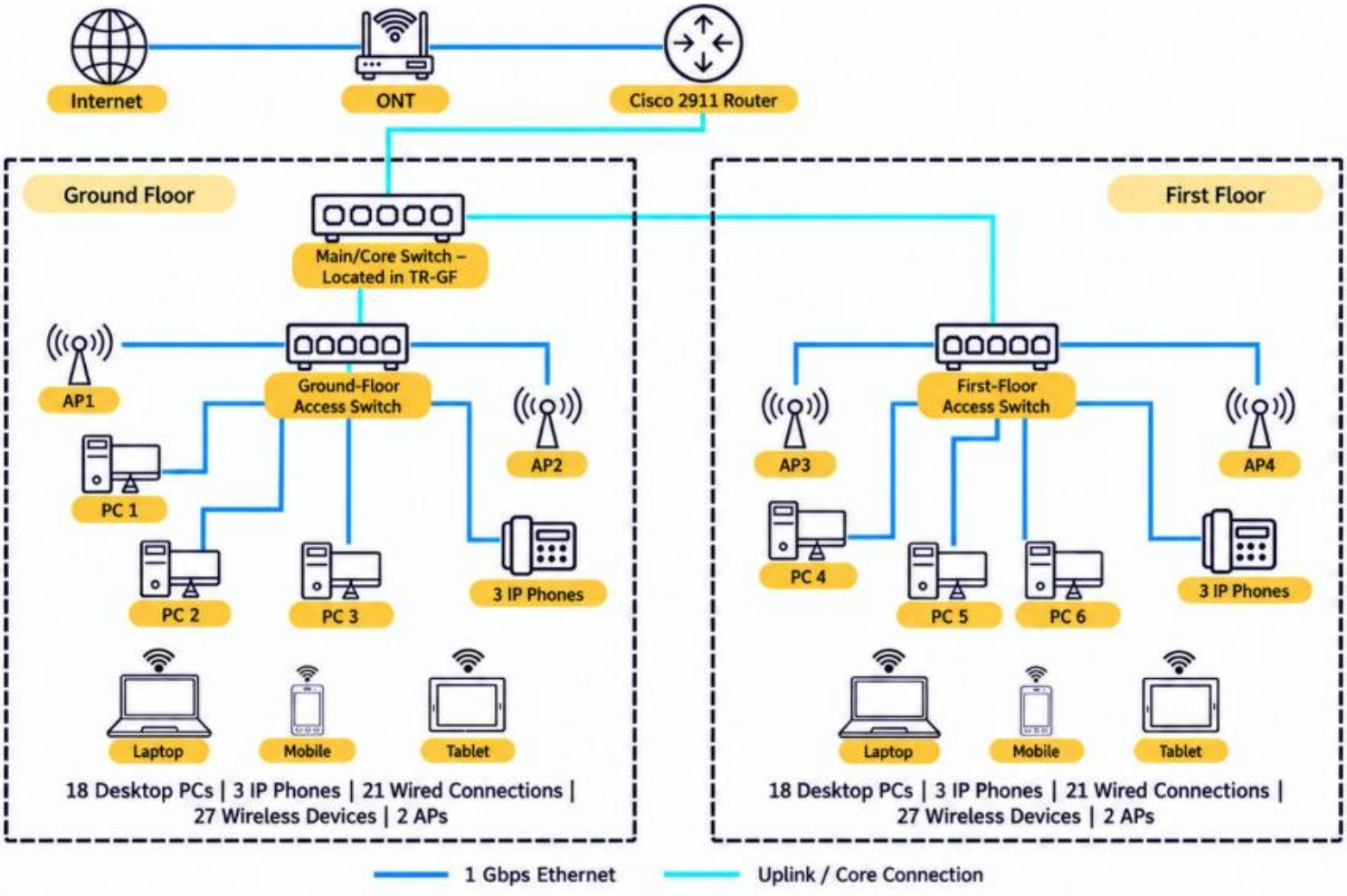
The use of high-speed links provides efficient communication between network devices and supports future network expansion

3.Physical Network Design

3.1 Physical Network Layout

The physical network design shows where network equipment, cabling, access points and network outlets are placed inside the two-story office building. Each floor has a network rack to keep cabling neat make it easy to maintain and allow for adding more equipment later.

The ground floor uses the TR-GF rack and the first floor uses the TR-FF rack. Each rack has a 24-port PoE switch, 24-port patch panel, UPS and cable management tools. The physical setup gives 42 wired network connections and 4 wireless access points with two APs on each floor.



3.2 Backbone and horizontal Cabling

The network uses two types of cabling, as summarised below. This structured cabling makes the network easier to manage and maintain.

Cabling	Medium	Route	Quantity
Horizontal	Cat6 UTP	Floor patch panel to RJ45 wall outlets and AP ceiling outlets	46 runs, 794 m including waste
Backbone	OM3 multimode fiber, 4-core	TR-GF fiber patch panel to TR-FF fiber patch panel	30 m ordered (about 22 m route)

- Horizontal Cabling: Cat6 UTP from the floor rack to network outlets and wired devices.
- Backbone Cabling: Fiber-optic cable connecting the ground-floor and first-floor network racks.

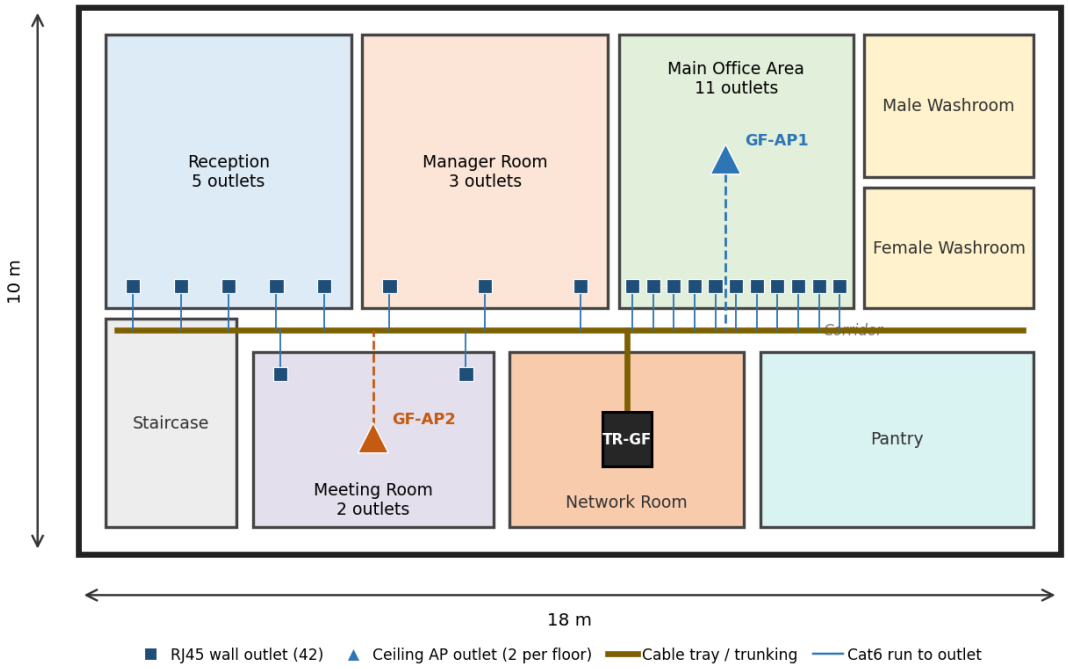
3.3 Cat6 UTP Cabling

Cat6 UTP is used for network connections to computers, IP phones and wireless access points. It connects the floor patch panel to RJ45 wall outlets. Supports a 1 Gbps Ethernet setup.

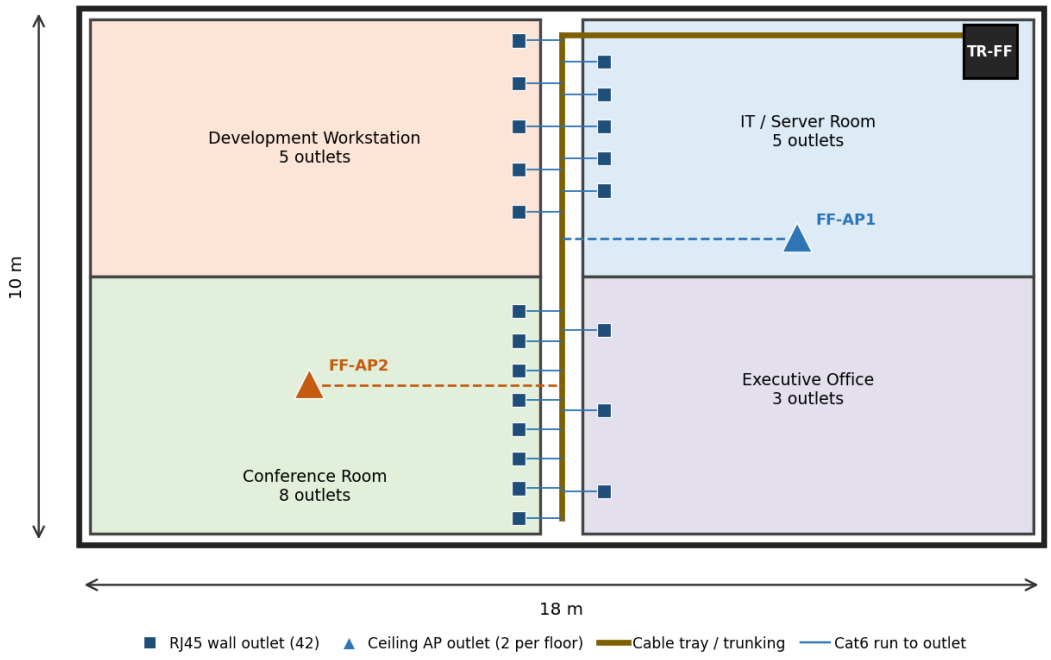
Cables are placed in the paths and clearly labeled for easy maintenance.

Cat6 UTP Cabling from Network Rack to User Outlets

Ground Floor - Outlets, Cat6 Cable Tray and Rack



First Floor - Outlets, Cat6 Cable Tray and Rack



3.4 Patch Panels and Wall Outlets

One 24-port Cat6 patch panel is installed in each floor rack. Each floor needs 23 ports (21 wired outlets + 2 access point outlets), so 23 of the 24 ports are used and 1 is spare. RJ45 wall outlets are placed on the wall within about 2 m of each desk, roughly 30 cm above the floor, with one outlet per desktop PC or IP phone. A short Cat6 patch cord connects each wall outlet to the device. The two access point outlets on each floor are surface-mounted at the ceiling position of each access point.

Wall Outlet Placement by Room

Floor	Room	Wall Outlets	AP Ceiling Outlet
Ground	Main Office Area	11	GF-AP1
Ground	Manager Room	3	-
Ground	Reception	5	-
Ground	Meeting Room	2	GF-AP2
Ground	**Subtotal**	**21**	**2**
First	IT / Server Room	5	FF-AP1
First	Development Workstation	5	-
First	Executive Office	3	-
First	Conference Room	8	FF-AP2
First	**Subtotal**	**21**	**2**
	Total	**42**	**4**

Patch Panel Port Usage

Floor	Wired Outlets	AP Outlets	Ports Needed	Panel Installed	Spare Ports
Ground floor (TR-GF)	21	2	23	1 x 24-port	1
First floor (TR-FF)	21	2	23	1 x 24-port	1
Total	**42**	**4**	**46**	**2 x 24-port**	**2**

Passive Components Summary

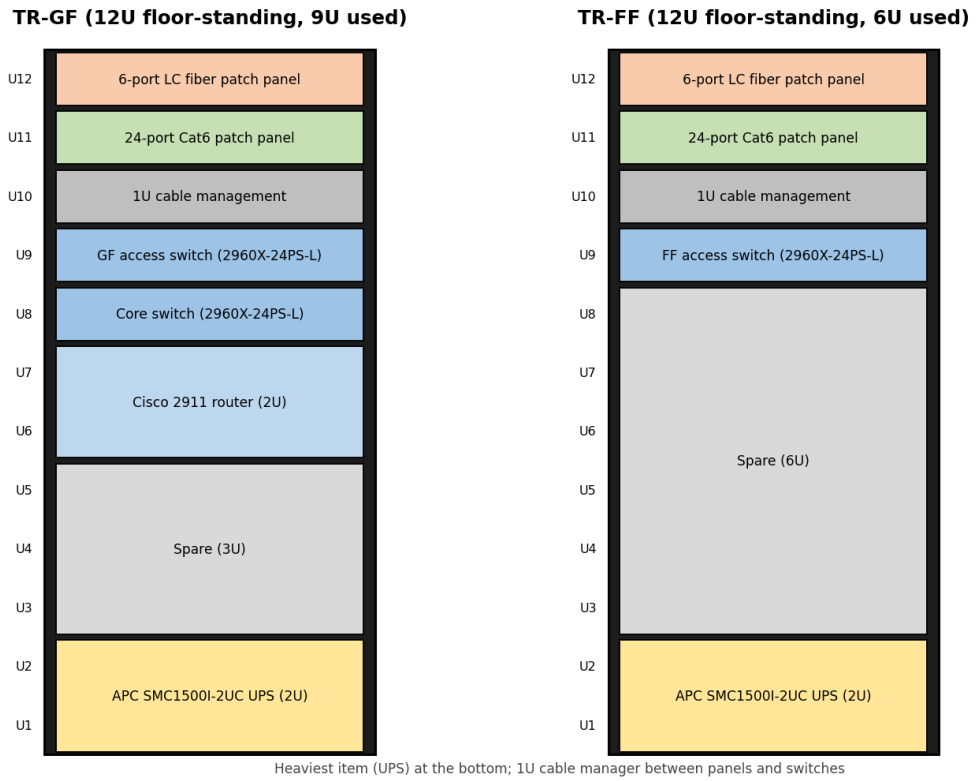
Component	Ground Floor	First Floor	Total
24-port Cat6 UTP patch panels	1	1	2
6-port LC fiber patch panels	1	1	2
RJ45 wall outlets	21	21	42
RJ45 ceiling outlets for APs	2	2	4
Cat6 patch cords (rack side, 1 m)	26 (23 + 3)	23	49
Cat6 patch cords (device side)	23 (21 x 2 m + 2 x 1 m)	23 (21 x 2 m + 2 x 1 m)	46
Spare Cat6 patch cords	-	-	5
LC-LC fiber patch cords	2	2	4

The 3 additional rack-side patch cords on the ground floor connect ONT to router, router to core switch, and core switch to GF-SW-01.

3.5 Equipment Rack Placement

Each floor has one 12U floor-standing rack (600 x 600 mm, glass door). The ground-floor rack TR-GF is in the Network Room beside the corridor, which keeps the cable tray short and gives easy access. The first-floor rack TR-FF is in the IT / Server Room, which is a secure room already used for equipment. A floor-standing rack is used because each rack carries a 1500 VA UPS, which is heavy and is mounted at the bottom of the rack.

Equipment	TR-GF	TR-FF
12U floor-standing network rack	1	1
Cisco 2911 router (2U)	1	0
Core switch, 24-port PoE (1U)	1	0
Access switch, 24-port PoE (1U)	1	1
24-port Cat6 patch panel (1U)	1	1
6-port LC fiber patch panel (1U)	1	1
Cable management unit (1U)	1	1
UPS, APC SMC1500I-2UC (2U)	1	1
Rack units used / spare	**9U / 3U**	**6U / 6U**



Heaviest item (UPS) at the bottom; 1U cable manager between panels and switches

3.6 Cable Pathing and Installation

Network cables are placed through cable trays, trunking, conduits and ceiling paths to keep the installation neat.

Important steps are:

- Keep data cables separate from power cables when possible.
- Avoid bends and too much pulling.
- Label cables, on both ends.
- Keep cabling organized inside the network racks.
- Leave some space for adding more later.

These steps help make a neat and easy-to-maintain network system.

4.Wireless Network Design

4.1 Access Point Placement and Coverage

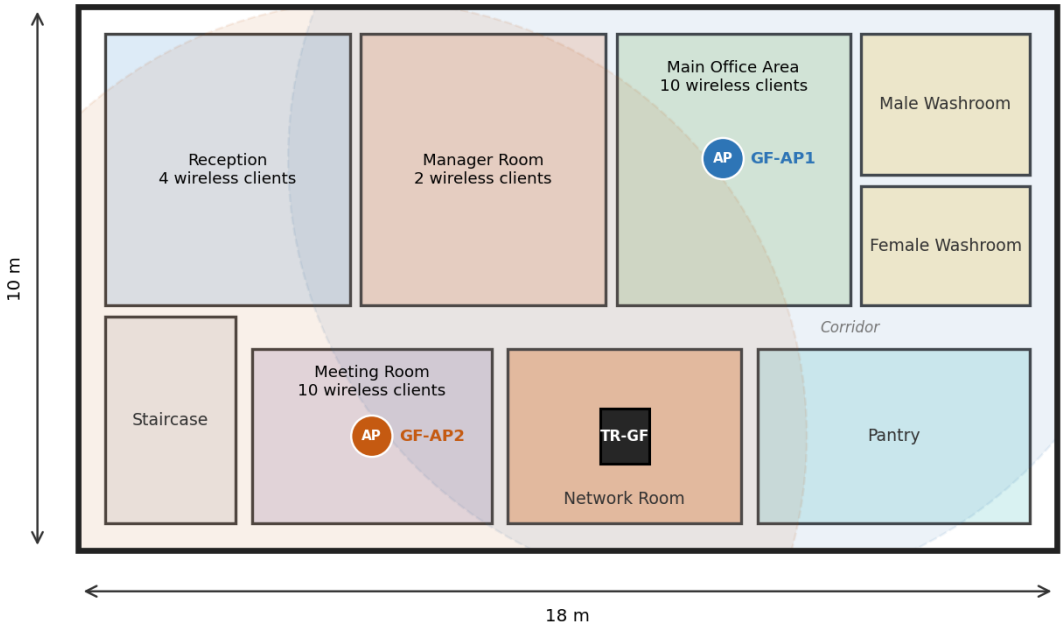
The wireless network has four access points, two on each floor, mounted on the ceiling. Each access point is placed in a central position between the two rooms it serves, so that the main working areas have strong coverage and weak zones are kept to a minimum. Each access point connects to its floor switch with Cat6 UTP through a ceiling outlet and a patch panel port.

AP	Location	Rooms Served	Wireless Clients
GF-AP1	Main Office Area	Main Office Area + Manager Room	12
GF-AP2	Meeting Room	Meeting Room + Reception	14
FF-AP1	IT / Server Room	IT / Server Room + Executive Office	12
FF-AP2	Conference Room	Conference Room + Development Workstation	16

Coverage justification. Each floor is 180 m² and has two access points, so each access point serves about 90 m². A planning radius of about 8 m is used, which is a conservative indoor figure for an office with internal partition walls. With this radius, every room on each floor lies inside the coverage of at least one access point. Each access point carries at most 16 clients, which is well within its capacity. A wireless survey should be done after installation to confirm signal levels and to fine-tune the channels (1, 6 and 11 on 2.4 GHz and non-overlapping channels on 5 GHz).

Ground Floor - Access Point Placement and Coverage

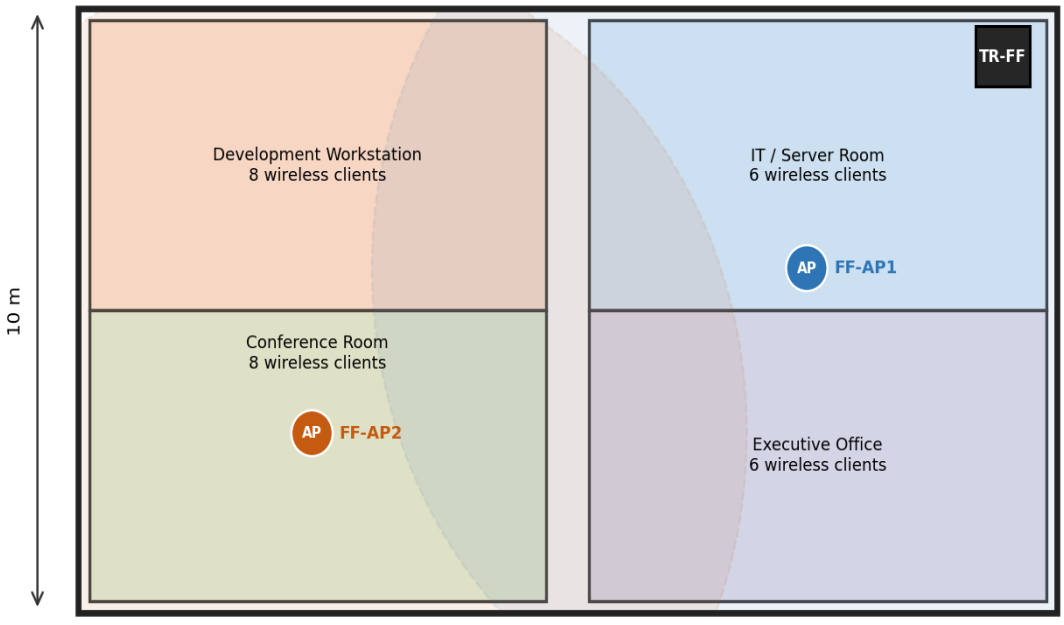
Dashed circle = ~8 m planning radius



- GF-AP1: Main Office Area + Manager Room
- GF-AP2: Meeting Room + Reception

First Floor - Access Point Placement and Coverage

Dashed circle = ~8 m planning radius



- FF-AP1: IT / Server Room + Executive Office
- FF-AP2: Conference Room + Development Workstation

4.2 Wireless Device Connectivity

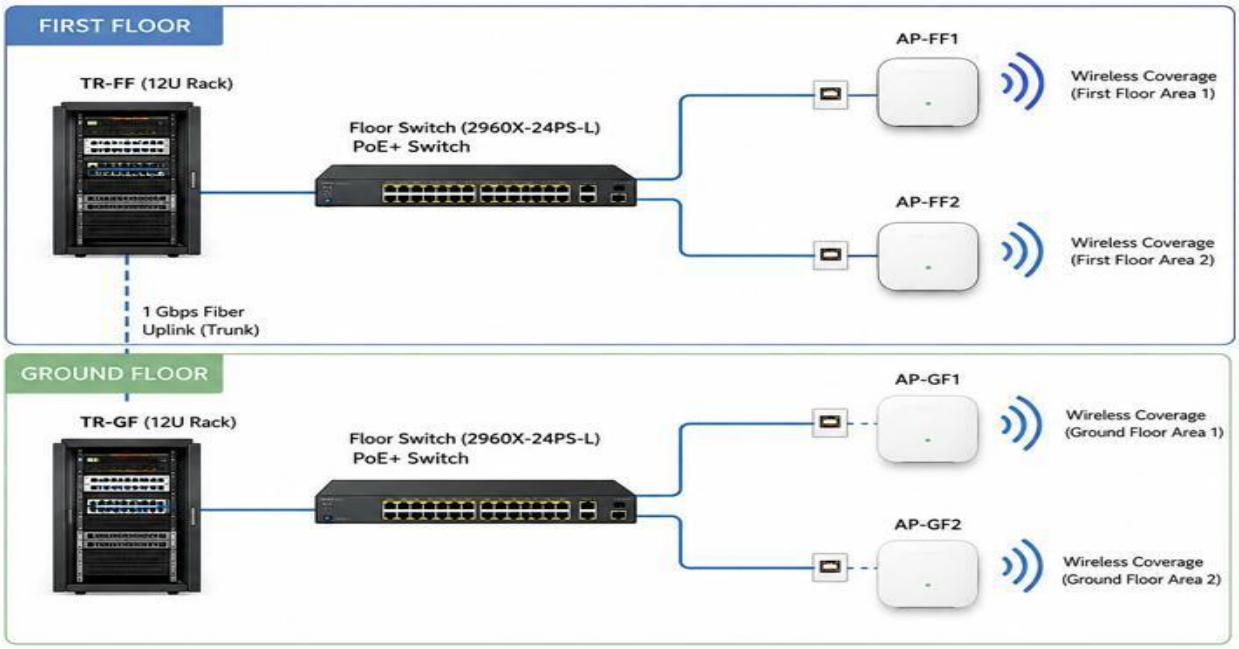
The wireless network is built to support up to 54 wireless client devices, which exceeds the requirement of 32 devices. The total number of devices expected in the office is higher than the minimum so the design accounts for future growth and usage demands

Wireless Device	Ground Floor	First Floor	Total
Laptops	11	14	25
Smartphones/Mobile Devices	15	14	29
Total Wireless Clients	**26**	**28**	**54**
Wireless APs	2	2	4

The access points deliver connectivity for laptops, smartphones and other mobile devices that employees use daily. This ensures everyone can stay connected regardless of location within the office.

4.3 AP Connectivity and Power Requirements

Each access point connects to the floor switch using Cat6 UTP Ethernet cable. Power is supplied through the network using PoE (Power over Ethernet) which means the access points receive both data and power through the cable. This reduces the need, for power outlets and simplifies the installation process. It also makes the network easier to maintain as there are components involved.



Wireless AP Connectivity through PoE Switches

5.Network Device Specification

5.1 Router and Network Switches

The Cisco 2911 Integrated Services Router (2U, 3 onboard Gigabit Ethernet ports) is the Internet gateway. It connects the office network to the Internet and handles routing between the internal network and the outside world.

Three Cisco Catalyst 2960X-24PS-L managed switches are used: one core switch and one access switch on each floor.

Specification	Cisco Catalyst 2960X-24PS-L
Quantity	3 (core switch in TR-GF, GF-SW-01 in TR-GF, FF-SW-01 in TR-FF)
Access ports	24 x 10/100/1000 Mbps copper, all PoE-capable (up to 12 ports at PoE+ at the same time)
Uplink ports	4 x 1 Gbps SFP
PoE budget	370 W per switch
Management	SNMP, IEEE 802.1Q VLAN support, LAN Base software
Form factor	1U, rack-mount

Switch Port Allocation

Switch	Copper Ports Used	SFP Ports Used	Spare Copper	Spare SFP
Core switch (TR-GF)	2 (router, GF-SW-01)	2 (FF fiber main + standby)	22	2
GF-SW-01 (TR-GF)	24 (23 outlets/APs + 1 uplink to core)	0	0	4
FF-SW-01 (TR-FF)	23 (21 outlets + 2 APs)	2 (fiber main + standby)	1	2

The core switch has 22 spare PoE-capable copper ports, which can serve extra devices on the ground floor in future.

5.2 Wireless Access Points

Specification	Proposed
Number of APs	4 (2 per floor)
Model	TP-Link EAP225 AC1350
Wireless standard	IEEE 802.11ac Wave 2 (MU-MIMO)
Frequency bands	2.4 GHz and 5 GHz
Wireless speed	Up to 450 Mbps (2.4 GHz) + 867 Mbps (5 GHz)
Network port	1 x Gigabit Ethernet
Cabling	Cat6 UTP
Power	PoE (IEEE 802.3af) from the floor switch
Mounting	Ceiling
Wireless clients	54 (maximum 16 per AP)

5.3 UPS and Equipment Racks

Two 12U floor-standing network racks are installed, one per floor. Each rack acts as the telecommunications rack (TR) of its floor and holds the switches, patch panels, cable management and UPS.

Equipment	Qty	Location	Model / Specification
12U network rack	2	1 per floor	Linkbasic 12U floor-standing, 600 x 600 mm, glass door
24-port PoE switch	3	2 in TR-GF, 1 in TR-FF	Cisco Catalyst 2960X-24PS-L
24-port Cat6 patch panel	2	1 per rack	Cat6 UTP, 24-port, 1U
6-port LC fiber patch panel	2	1 per rack	6-port LC, 1U
Cable management unit	2	1 per rack	1U 19-inch cable organizer
UPS	2	1 per rack	APC Smart-UPS SMC1500I-2UC (1500 VA / 900 W, 2U rack)
SFP module	4	2 in core switch, 2 in FF-SW-01	1000BASE-SX multimode

The ground-floor rack is labelled TR-GF and the first-floor rack TR-FF. These labels help with identification, troubleshooting and maintenance.

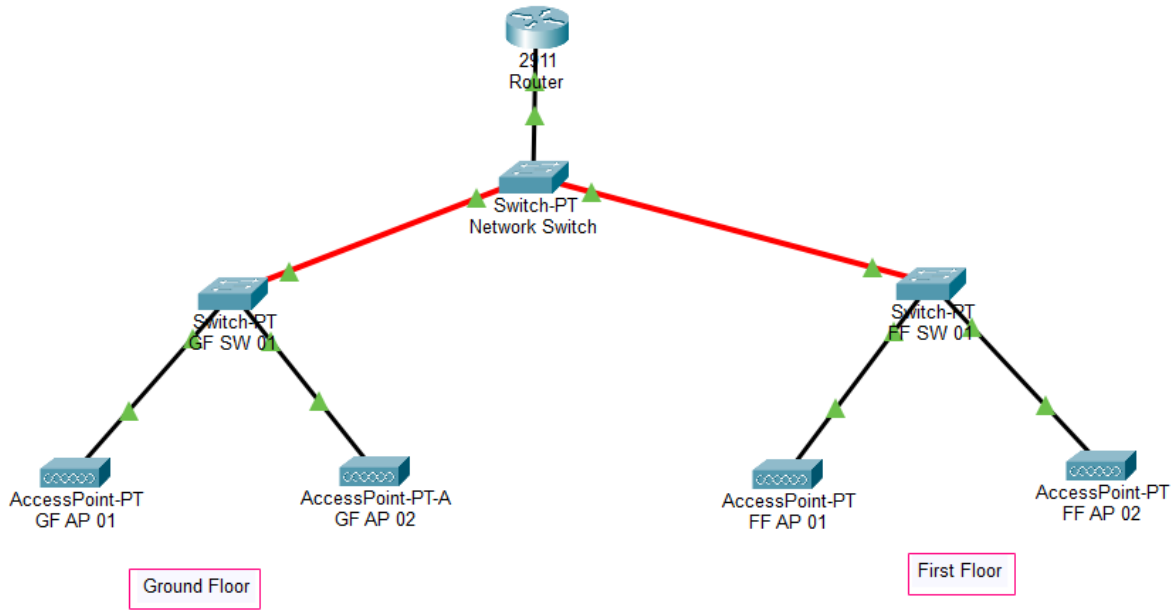
5.4 Patch Panel and Network Accessories

Component	Quantity
24-port Cat6 patch panels	2
6-port LC fiber patch panels	2
RJ45 outlets (42 wall + 4 ceiling AP)	46
Cat6 patch cords (1 m and 2 m)	100
LC-LC duplex fiber patch cords	4
1000BASE-SX SFP modules	4
Cat6 UTP cable boxes (305 m)	3 boxes (915 m)
Fiber optic backbone cable	30 m, 4-core OM3
Cable management units	2

6.Network Implementation and Cisco Packet Tracer

6.1 Packet Tracer Topology

The proposed network was implemented in Cisco Packet Tracer using one 2911 router, three 24-port switches, and four wireless access points.



Cisco Packet Tracer Network Topology Diagram

6.2 Device Interconnections

The Cisco 2911 router acts as the gateway and connects to the network switches. Each floor has one 24-port switch, which connects to the APs and wired network infrastructure.

Connection	Cable/Link	Design Speed
Router – Main Switch	Cat6 UTP Ethernet	1 Gbps
Main Switch – Ground Floor Switch	Fiber Backbone	1 Gbps
Main Switch – First Floor Switch	Fiber Backbone	1 Gbps
Ground Floor Switch – APs	Cat6 UTP	1 Gbps
First Floor Switch – APs	Cat6 UTP	1 Gbps

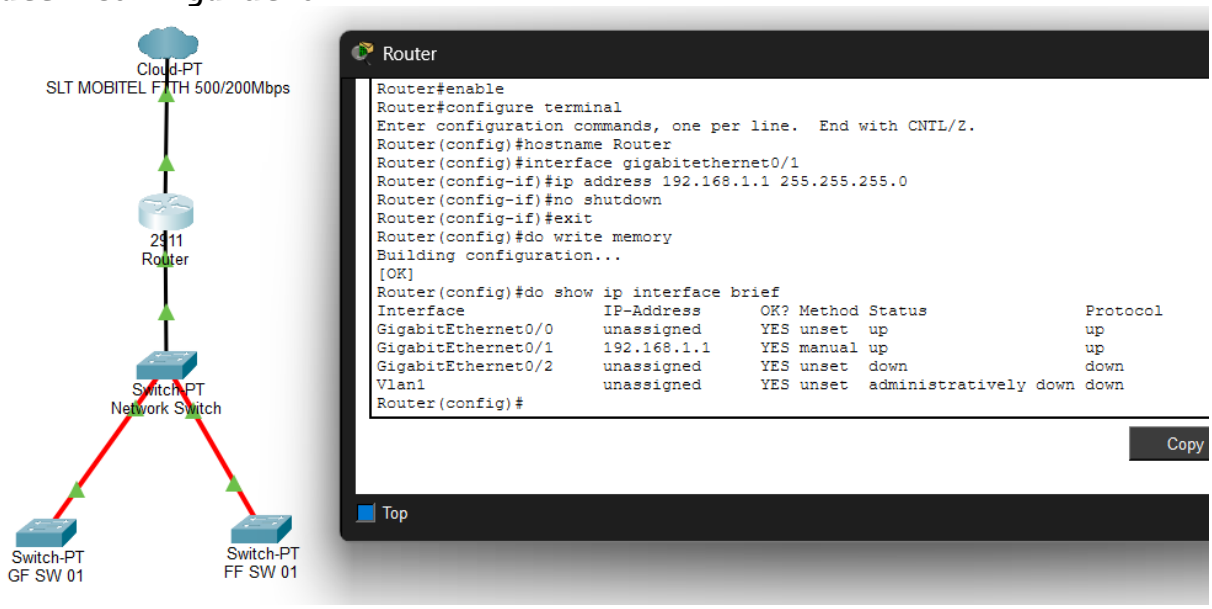
6.3 Routers, Switch and Interface Configuration

The Cisco 2911 router and network switches are configured using the Cisco IOS interface. Appropriate interfaces are. Assigned according to the proposed network design.

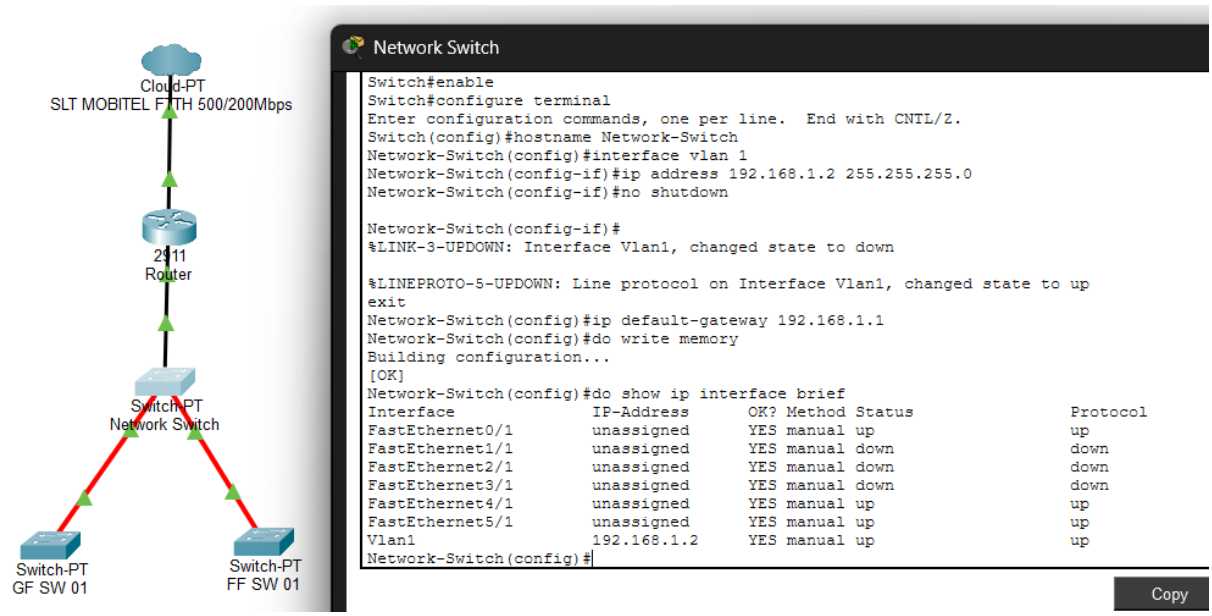
Basic configuration includes:

- Device hostnames
- Interface activation
- IP addressing
- Default gateway configuration
- Switch port configuration
- Appropriate interface descriptions

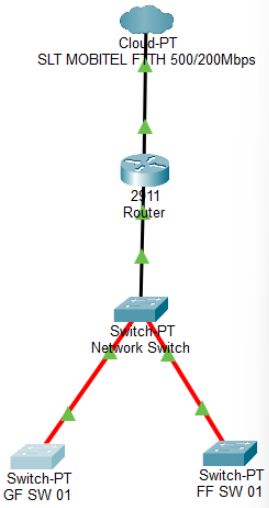
Router Configuration



Network Switch Configuration



Ground Floor Switch Configuration



GF SW 01

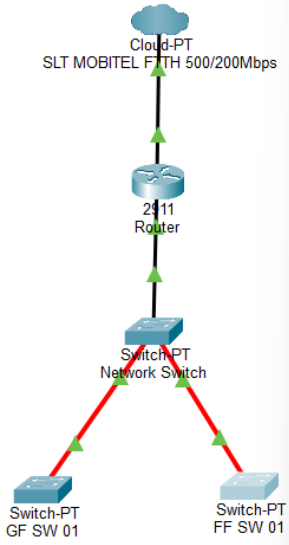
Physical Config CLI Attributes

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname GF-SW-01
GF-SW-01(config)#interface vlan 1
GF-SW-01(config-if)#ip address 192.168.1.3 255.255.255.0
GF-SW-01(config-if)#no shutdown

GF-SW-01(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up
exit
GF-SW-01(config)#ip default-gateway 192.168.1.1
GF-SW-01(config)#do write memory
Building configuration...
[OK]
GF-SW-01(config)#do show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/1        unassigned      YES manual down                  down
FastEthernet1/1        unassigned      YES manual down                  down
FastEthernet2/1        unassigned      YES manual down                  down
FastEthernet3/1        unassigned      YES manual down                  down
FastEthernet4/1        unassigned      YES manual up                    up
FastEthernet5/1        unassigned      YES manual down                  down
GigabitEthernet6/1     unassigned      YES manual down                  down
Vlan1                  192.168.1.3     YES manual up                    up
GF-SW-01(config)#
```

First Floor Switch Configuration



FF SW 01

Physical Config CLI Attributes

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname FF-SW-01
FF-SW-01(config)#interface vlan 1
FF-SW-01(config-if)#ip address 192.168.1.4 255.255.255.0
FF-SW-01(config-if)#no shutdown

FF-SW-01(config-if)#
%LINK-3-UPDOWN: Interface Vlan1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up
exit
FF-SW-01(config)#ip default-gateway 192.168.1.1
FF-SW-01(config)#do write memory
Building configuration...
[OK]
FF-SW-01(config)#do show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/1        unassigned      YES manual down                  down
FastEthernet1/1        unassigned      YES manual down                  down
FastEthernet2/1        unassigned      YES manual down                  down
FastEthernet3/1        unassigned      YES manual down                  down
FastEthernet4/1        unassigned      YES manual up                    up
FastEthernet5/1        unassigned      YES manual down                  down
GigabitEthernet6/1     unassigned      YES manual down                  down
Vlan1                  192.168.1.4     YES manual up                    up
FF-SW-01(config)#
```

Copy

6.4 Link Speed Configuration

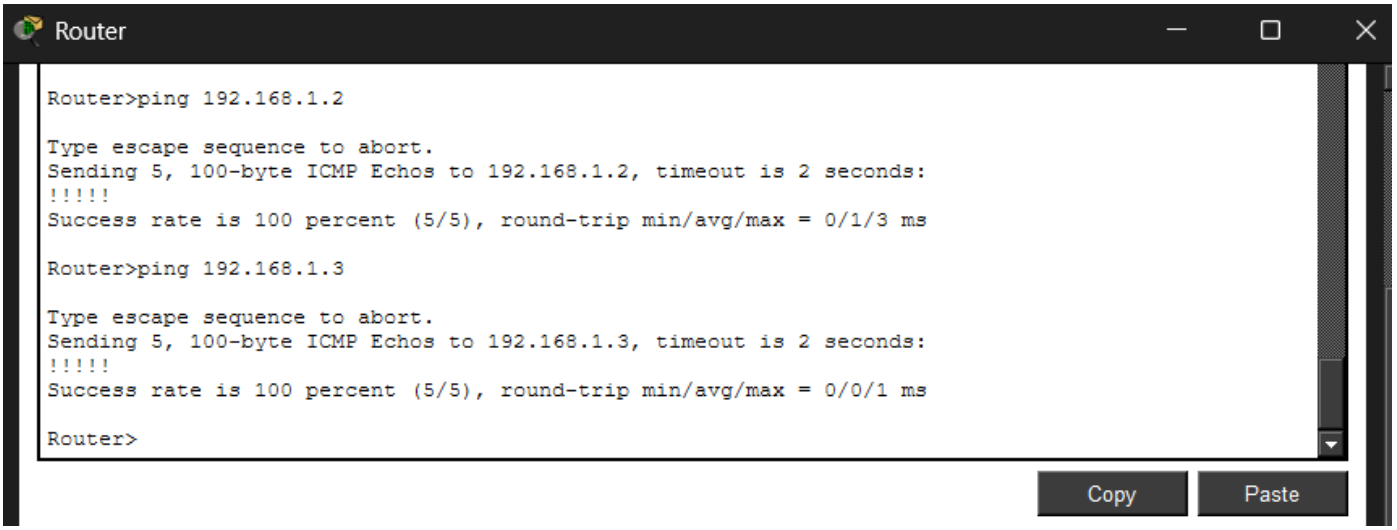
The main network links are designed for 1 Gbps. Interface speed and status can be verified using:

show interfaces

The output can be used to confirm interface status, speed and duplex settings.

6.5 Network Connectivity Testing

Connectivity is tested after configuring the Cisco 2911 router the network switches and the APs. The ping command is used to verify connectivity, between network devices. Successful responses confirm that the configured network paths are operating correctly.



7.Cable and Port Calculations

7.1 Wired Port and Patch Panel Requirements

The new network needs 42 wired network connections, which includes 36 desktop PCs and 6 IP phones.

Floor	Wired Connections	Patch Panel
Ground Floor	21	1 × 24-port
First Floor	21	1 × 24-port
Total	**42**	**2 × 24-port**

Every floor has patch-panel ports for its current needs with more ports available for future use.

7.2 Patch Cord Requirements

Category	Description	Quantity
Rack-side Cat6 cords (1 m)	46 patch-panel-to-switch cords (23 per floor) + 3 equipment links (ONT-router, router-core switch, core switch-GF-SW-01)	49
Device-side Cat6 cords (2 m)	36 desktop PCs + 6 IP phones, one per wall outlet	42
AP-side Cat6 cords (1 m)	Ceiling outlet to access point, 1 per AP	4
Spare Cat6 cords	About 5 percent spares for maintenance	5
Total Cat6 patch cords		**100**
LC-LC fiber patch cords (2 m)	2 links (main + standby) x 2 ends, OM3 duplex 50/125 um	4

Cat6 patch cords are factory-moulded with RJ45 connectors. The LC-LC cords are duplex multimode OM3 (50/125 um) cords.

7.3 Cat6 Cable Length Estimation

The estimate uses an average run of 15 m per drop. This covers about 9 m of horizontal route along the cable tray, about 5 m of vertical rise at both ends (rack and outlet), and about 1 m of slack. Wired outlets and access point outlets are both counted, and 15 percent is added for waste and routing.

Floor	Drops (wired + AP)	Average Length	Cable Needed
Ground floor	21 + 2 = 23	15 m	345 m
First floor	21 + 2 = 23	15 m	345 m
Subtotal	**46**	**-**	**690 m**
Waste and routing allowance (15%)			103.5 m
Total cable required			**794 m**

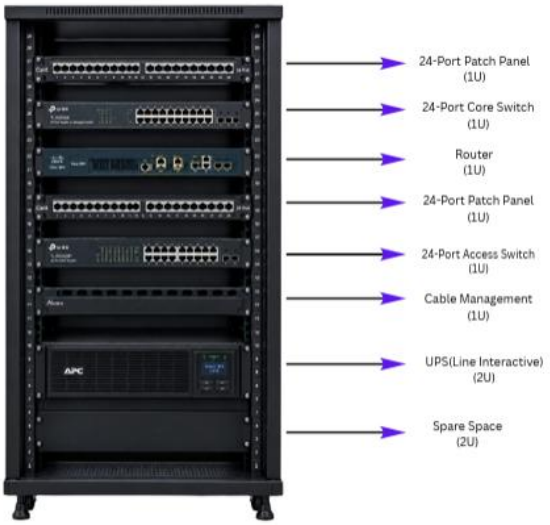
794 m of Cat6 UTP cable is required, so 3 boxes of 305 m (915 m) are purchased, leaving about 121 m spare.

Fiber backbone length. The route is about 5 m on the ground floor to the riser, about 4 m vertically between floors, and about 13 m on the first floor to TR-FF, which is about 22 m. With 20 percent slack for service loops this is about 26.4 m, so a 30 m 4-core OM3 cable is ordered.

7.4 Equipment Rack Requirements

Two 12U racks are needed, one per floor. Both have spare rack units for future growth.

Rack	Floor	Size	Main Equipment	Used / Spare
TR-GF	Ground floor	12U	Router, core switch, access switch, Cat6 and fiber patch panels, cable manager, UPS	9U / 3U
TR-FF	First floor	12U	Access switch, Cat6 and fiber patch panels, cable manager, UPS	6U / 6U



U	Equipment
1U	24-Port Patch Panel
1U	24-Port Core Switch
1U	Router
1U	24-Port Patch Panel
1U	24-Port Access Switch
1U	Cable Management
2U	UPS(Line Interactive)
2U	Spare Space

8.Detailed Cost Estimate

The cost estimate covers network equipment, passive components, cabling, installation and Internet service. Prices are based on Sri Lankan supplier quotations or official service information available at the time of submission. The SFP module price is an estimate and should be confirmed with a vendor quotation.

8.1 Active Network Equipment

Item	Qty	Unit Cost (LKR)	Total (LKR)	Pricing Source / Reference
Cisco 2911 Integrated Services Router	1	288,723	288,723	Cisco LK Authorized Dealer
Cisco Catalyst 2960X-24PS-L 24-port PoE switch	3	86,250	258,750	Barclays Computers / Vendor List
TP-Link EAP225 AC1350 Wireless AP	4	30,500	122,000	Nanotek IT Solutions / TP-Link LK
APC Smart-UPS SMC1500I-2UC (1500VA)	2	70,000	140,000	Singer LK / Barclays Computers
1000BASE-SX multimode SFP module	4	12,000	48,000	Estimate - confirm with vendor quotation
Sub Total			**857,473**	

8.2 Passive Equipment and Cabling

Item	Qty	Unit Cost (LKR)	Total (LKR)	Pricing Source / Reference
Cat6 UTP cable box (305 m)	3	48,000	144,000	Schneider / DINTEK LK Official
24-port Cat6 patch panel	2	10,750	21,500	DINTEK Sri Lanka
6-port LC fiber patch panel	2	14,500	29,000	Fiber optic local distributor
RJ45 outlet (42 wall + 4 ceiling AP)	46	1,500	69,000	DINTEK / Local Hardware Depot
Cat6 patch cords (1 m / 2 m)	100	1,200	120,000	DINTEK Factory Molded Cords
LC-LC duplex fiber patch cords	4	3,500	14,000	Local Fiber Supplier Quote
12U floor-standing network rack	2	48,500	97,000	Linkbasic Sri Lanka
Fiber optic backbone, 30 m 4-core OM3 (supply and termination)	1	25,000	25,000	Indoor Fiber Supplier
1U cable management set	2	6,500	13,000	Linkbasic Accessories
Sub Total			**532,500**	

8.3 Installation and Labor

Installation costs include cable routing, termination, rack installation, outlet installation, testing and configuration.

Service	Quantity	Cost (LKR)	Pricing Source
Network installation and labor	1	100,000	Certified Network Contractor
Cable termination and testing	1	30,000	Fluke Testing Service Contractor
Subtotal		**130,000**	

8.4 Internet Service Cost

Requirement	Proposed Solution
ISP	SLT-MOBITEL
Connection	Fiber Optic
Package	Fiber Unlimited Pro
Speed	500 Mbps download / 200 Mbps upload
Monthly cost	Rs. 19,900 + taxes

Note: Including Sri Lankan telecommunication taxes and SSCL / VAT (about 30 percent), the estimated gross monthly Internet cost is approximately LKR 25,870.

8.5 Total Project Cost

Cost Category	Amount (LKR)
Active network equipment	857,473
Passive equipment and cabling	532,500
Installation and labor	130,000
Initial project cost	**1,519,973**
First month Internet cost	25,870
Total including first month Internet	**1,545,843**

9.Network Verification and Evaluation

9.1 Connectivity and Performance Verification

The proposed network is verified using Cisco Packet Tracer and basic connectivity tests. Ping tests are used to confirm communication between network devices.

The following aspects are checked:

- Router and switch connectivity
- Inter-floor switch connectivity
- AP connectivity
- Interface status
- Link speed and duplex settings
- End-device communication

Successful tests indicate that the logical network configuration operates as intended.

9.2 Physical Design Verification

The physical design was checked against the floor layouts and the equipment requirements. The figures used in every section of this report were cross-checked and agree with each other.

Check	Result
Outlets: 42 wall + 4 AP = 46 cabled outlets	= 46 patch panel ports (23 per floor), 2 x 24-port panels with 1 spare each
Outlets per room match the room tables	GF 21 and FF 21 wired drops
Wireless clients per AP	12, 14, 12 and 16, none above 16
Rack space	TR-GF 9U of 12U, TR-FF 6U of 12U
Cat6 cable	794 m required, 915 m purchased
Patch cords	100 Cat6 cords and 4 LC-LC cords, same figure in every section
PoE budget	About 72 W worst case per floor against 370 W available
Cable routes	Tray along the corridor, data cables kept away from power cables

9.3 Requirement Compliance

The final design satisfies the network requirements of the proposed office.

Requirement	**Proposed Design**	**Status**
More than 32 wired devices	42 wired connections	Satisfied
More than 32 wireless devices	54 wireless clients	Satisfied
Wireless APs	4 APs	Satisfied
Floor switches	3 × 24-port PoE switches	Satisfied
Patch panels	2× 24-port	Satisfied
Network racks	2 × 12U	Satisfied
Wired cabling	Cat6 UTP	Satisfied
Inter-floor backbone	Fiber optic	Satisfied
Wired design speed	1 Gbps	Satisfied
Internet connectivity	SLT-MOBITEL Fiber	Satisfied

The verification confirms that the proposed network design meets the identified physical logical wired, wireless and connectivity requirements.

8.Conclusion and Recommendations

The proposed network provides a scalable and well-structured infrastructure for the two-story office building. It supports 42 wired connections and 54 wireless clients using four wireless access points. It uses Cat6 UTP for cabling and fiber optic cabling for the inter-floor backbone. The design supports the required 1 Gbps connectivity while racks, patch panels, UPS units and organized cabling improve reliability and maintenance.

Recommendations:

• Maintain cable labeling and documentation.

• Reserve spare ports, for expansion.

• Perform network, UPS and cable testing.

• Monitor wireless. Adjust AP placement if required.

• Update device. Upgrade equipment as network requirements increase

References

1. Cisco Systems. (2026). Cisco 2911 Integrated Services Router – Product Documentation. Cisco Systems.

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5. IEEE. (2026). IEEE 802.3 Ethernet Standards. Institute of Electrical and Electronics Engineers.

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7. Cisco Networking Academy. (2026). Cisco Packet Tracer – Network Simulation and Design Resources. Cisco.

Group Member Contribution

IT Number	Group Member	Requirement Analysis (20%)	Logical Network Design (25%)	Physical Network Design (25%)	Device Specifications (15%)	Cost Estimation (15%)	**Overall Contribution**
IT26102361	**VINOJA H.W.O.** (Team Leader)	3.33%	4.17%	4.17%	2.50%	2.50%	**16.67%**
IT26102355	**KULATHUNGA M.A.S.S.**	3.33%	4.17%	4.17%	2.50%	2.50%	**16.67%**
IT26102368	**PERERA M.K.J.C.**	3.33%	4.17%	4.17%	2.50%	2.50%	**16.67%**
IT26102357	**SUBHASHANA T.O.**	3.33%	4.17%	4.17%	2.50%	2.50%	**16.67%**
IT26102317	**SITHUMINI W.T.I.**	3.33%	4.17%	4.17%	2.50%	2.50%	**16.67%**
IT26102348	**DAMAYANTHA H.A.E.**	3.35%	4.15%	4.15%	2.50%	2.50%	**16.65%**
TOTAL		**20.00%**	**25.00%**	**25.00%**	**15.00%**	**15.00%**	**100.00%**